

# Bearings and angles of elevation in three dimensions

Cambridge insert: navigation and surveying, where the plan view and the vertical triangle are solved separately.

LESSON 48–49

2 × 40 MIN

GEOMETRY 10, §18 (EXTENSION)

IGCSE E14.2 · E6.6

LESSON CLOCK

12'

22'

26'

22'

20'

8'

|                           |        |
|---------------------------|--------|
| Bearings                  | 12 min |
| Elevation and depression  | 22 min |
| Working in the plan first | 26 min |
| Combined problems         | 22 min |
| Practice                  | 20 min |
| Homework                  | 8 min  |

LEARNING OBJECTIVES

- Measure and use a three-figure bearing.
- Find the angle of elevation and depression in a 3-D setting.
- Solve a navigation problem by working in the plan first.
- Combine the plan triangle with the vertical triangle.

TEXTBOOK

|           |                                       |
|-----------|---------------------------------------|
| National  | pp. 229–234                           |
| Cambridge | Core & Extended, pp. 217–222, 305–312 |
| Workbook  | IGCSE Exercise 14.2                   |

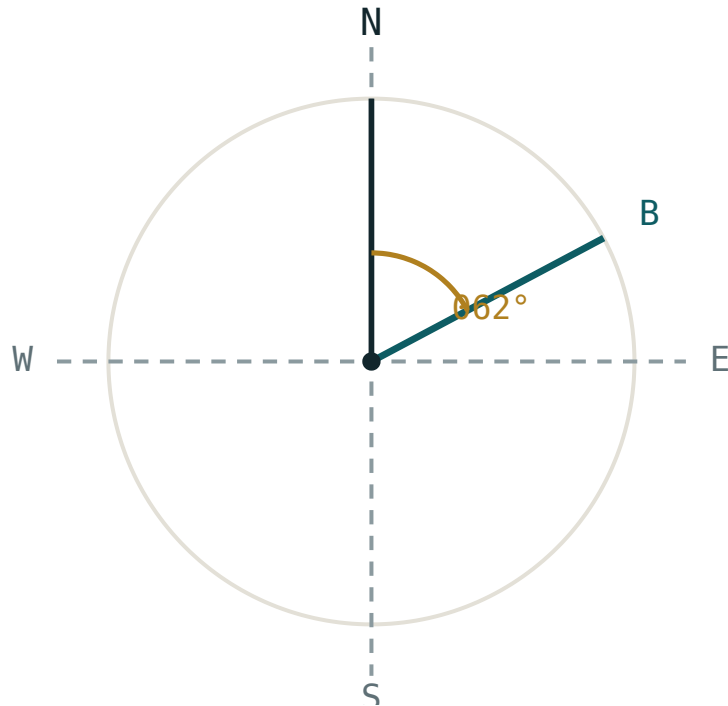
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## Explanation

### Bearings

THE CONVENTION

A **bearing** is measured **from north, clockwise**, and written with **three figures**:  $045^\circ$ ,  $120^\circ$ ,  $300^\circ$ . Due east is  $090^\circ$ , south  $180^\circ$ , west  $270^\circ$ .



*North at the top, clockwise, always three figures.*

The **back bearing** — the bearing of  $A$  from  $B$  when the bearing of  $B$  from  $A$  is  $\theta$  — is  $\theta + 180^\circ$ , reduced by  $360^\circ$  if it exceeds it.

#### THREE FIGURES, ALWAYS

$45^\circ$  is written  $045^\circ$ . It is not decoration: on a chart a missing leading zero has caused real navigational error.

## Elevation and depression

The **angle of elevation** of a point is the angle its line of sight makes with the horizontal, measured upwards; the **angle of depression** is the same measured downwards.

angle of depression from  $A$  to  $B$  = angle of elevation from  $B$  to  $A$

They are alternate angles between two horizontals — which is why a question may give either and expect the other.

## Working in the plan first

THE METHOD FOR EVERY 3-D NAVIGATION PROBLEM

- 1** Draw the **plan** — a flat map — and solve it with the sine or cosine rule to find the horizontal distance.
- 2** Draw the **vertical triangle** containing that horizontal distance and the height. **3** Solve it with ordinary right-angle trigonometry.

**Example.** From  $A$ , a tower is on a bearing of  $070^\circ$  at  $300$  m; from  $B$ ,  $400$  m due east of  $A$ , its bearing is  $340^\circ$ . The angle of elevation of the top from  $A$  is  $9^\circ$ . Find its height.

The plan gives the horizontal distance  $AT = 300$  m directly, so the vertical triangle gives  $h = 300 \tan 9^\circ \approx 47.5$  m. The second bearing is a check.

## Combined problems

A full surveying problem chains three steps:

1. bearings  $\rightarrow$  a plan triangle  $\rightarrow$  the cosine or sine rule  $\rightarrow$  the horizontal distance;
2. horizontal distance + angle of elevation  $\rightarrow$  the height;
3. height + a second horizontal distance  $\rightarrow$  a second angle.

**Example.** A mast stands at  $C$ . From  $A$  its elevation is  $12^\circ$ ; from  $B$ ,  $200$  m from  $A$  on a bearing making  $50^\circ$  at  $C$  with  $AC$ , its elevation is  $18^\circ$ . With  $h$  the height,  $AC = h \cot 12^\circ$  and  $BC = h \cot 18^\circ$ ; the cosine rule in the plan then gives one equation in  $h$  alone.

### NEVER MIX THE PLAN AND THE VERTICAL TRIANGLE

Bearings live in the plan; elevations live in the vertical. Solve one, carry the single number across, and solve the other. Trying to do both at once is where every mark is lost.

## 02 Worked examples

**EXAMPLE 1** The bearing of  $B$  from  $A$  is  $062^\circ$ . Find the bearing of  $A$  from  $B$ .

①  $62 + 180 = 242$

② Less than  $360^\circ$ , so no reduction.

ANSWER

$242^\circ$

**EXAMPLE 2** A tower is 300 m away horizontally; its top has elevation  $9^\circ$ . Find its height.

$$\textcircled{1} \quad \tan 9^\circ = \frac{h}{300}$$

$$\textcircled{2} \quad h = 300 \times 0.1584$$

$$\textcircled{3} \quad \approx 47.5 \text{ m.}$$

ANSWER

$\approx 47.5 \text{ m}$

**EXAMPLE 3** From a cliff 80 m high, a boat has an angle of depression of  $23^\circ$ . Find its distance from the foot.

$\textcircled{1}$  The angle of elevation from the boat is also  $23^\circ$ .

$$\textcircled{2} \quad \tan 23^\circ = \frac{80}{d}$$

$$\textcircled{3} \quad d = \frac{80}{0.4245} \approx 188.4 \text{ m.}$$

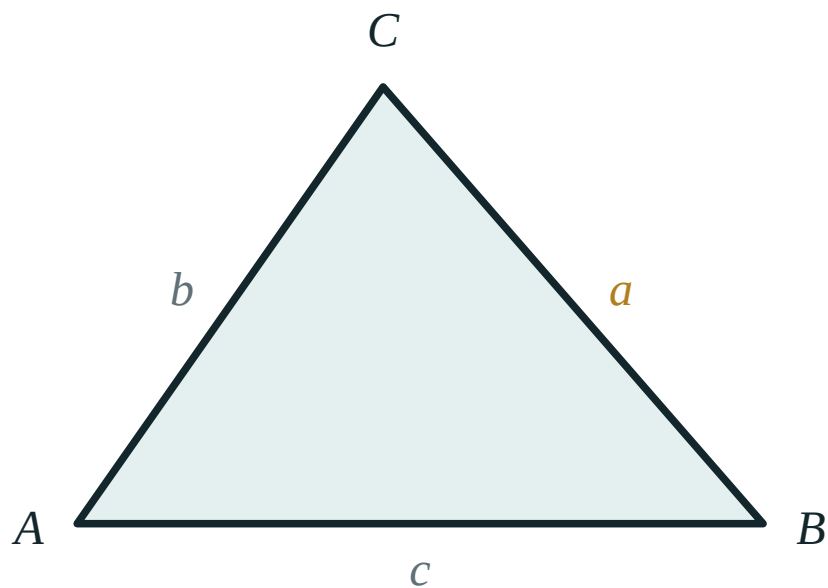
ANSWER

$\approx 188 \text{ m}$

### 03 Interactive model

Draw the plan on one half of the board and the vertical triangle on the other.

## The plan triangle



b 7

c 9

A  $55^\circ$ 

a 7.6

B  $49^\circ$ C  $76^\circ$ 

area 25.8

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH              | O'ZBEKCHA            | РУССКИЙ            |
|----------------------|----------------------|--------------------|
| Bearing              | Azimut               | Азимут             |
| Three-figure bearing | Uch xonali azimuth   | Трёхзначный азимут |
| Angle of elevation   | Ko'tarilish burchagi | Угол возвышения    |
| Angle of depression  | Pasayish burchagi    | Угол понижения     |
| Plan view            | Yuqoridan ko'rinish  | Вид сверху         |
| Due north            | Shimolga             | На север           |
| Back bearing         | Teskari azimuth      | Обратный азимут    |
| Line of sight        | Ko'rish chizig'i     | Линия визирования  |
| Surveying            | Geodeziya            | Геодезия           |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

A bearing is measured from:

east, anticlockwise

north, clockwise

north, anticlockwise

south, clockwise

Due west is:

$090^\circ$

$180^\circ$

$270^\circ$

$360^\circ$

The back bearing of  $062^\circ$  is:

$118^\circ$

$242^\circ$

$298^\circ$

$062^\circ$

The angle of depression from  $A$  to  $B$  equals:

the elevation from  $B$  to  $A$

its complement

nothing

twice it

In a 3-D navigation problem, solve first:

the vertical triangle

the plan

both at once

neither

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

**Easy · warm the idea up**

7 PROBLEMS



1. Bearing of due east

ANSWER  $090^\circ$

2. Bearing of due south

ANSWER  $180^\circ$

3. Bearing of due west

ANSWER  $270^\circ$

4. Back bearing of  $045^\circ$

ANSWER  $225^\circ$

5. Back bearing of  $300^\circ$

ANSWER  $120^\circ$

6. Height with distance 100, elevation  $30^\circ$

ANSWER  $\approx 57.7$

7. Distance with height 50, depression  $45^\circ$

ANSWER 50

Medium · the standard question

7 PROBLEMS

1. Back bearing of  $062^\circ$

ANSWER  $242^\circ$

2. Tower 300 m away, elevation  $9^\circ$ : its height

ANSWER  $\approx 47.5$  m

3. Cliff 80 m, depression  $23^\circ$ : the distance

ANSWER  $\approx 188$  m

4. A mast of 40 m at 120 m: the angle of elevation

ANSWER  $\approx 18.4^\circ$

5. A plane at 3000 m, depression to the airport  $12^\circ$ : the ground distance

ANSWER  $\approx 14\ 100$  m

6. Bearing  $130^\circ$  for 5 km: how far east and how far south?

ANSWER 3.83 km east, 3.21 km south

7. From A, B is at  $070^\circ$  4 km; C at  $160^\circ$  3 km. Find BC

ANSWER 5 km

Hard · stretch and reason

7 PROBLEMS

1. A tower: elevations  $12^\circ$  and  $18^\circ$  from two points 200 m apart in line with it. Find its height

ANSWER  $\approx 79.9$  m

2. Same, with the two points and the tower forming a plan angle of  $50^\circ$

ANSWER Cosine rule in  $h \cot 12^\circ$  and  $h \cot 18^\circ$

3. A ship sails  $070^\circ$  for 8 km then  $160^\circ$  for 6 km. Find its distance and bearing from the start

ANSWER 10 km on  $\approx 107^\circ$

4. From a hill 120 m high, two boats have depressions  $15^\circ$  and  $25^\circ$  on the same bearing. Distance apart

ANSWER  $\approx 190$  m

5. A mountain has elevation  $8^\circ$  from 20 km and  $11^\circ$  from a nearer point. Find the height and the second distance

ANSWER  $\approx 2.81$  km;  $\approx 14.5$  km

6. A pylon is at  $040^\circ$ , 500 m from A and at  $310^\circ$  from B due east of A. Find AB

ANSWER  $\approx 780$  m

7. Explain why the plan is solved first

ANSWER It supplies the single horizontal distance the vertical triangle needs

## 07 Homework

### Homework — 5 tasks

Draw the plan and the vertical triangle as two separate diagrams.

1. Give the back bearings of  $035^\circ$ ,  $145^\circ$  and  $290^\circ$ .
2. A tower is 450 m away and its top has elevation  $11^\circ$ . Find its height.
3. From a cliff 95 m high a boat has depression  $19^\circ$ . Find its distance from the foot.
4. A ship sails  $060^\circ$  for 12 km, then  $150^\circ$  for 9 km. Find its distance and bearing from the start.

5. A mast: from two points  $150$  m apart in line with it the elevations are  $14^\circ$  and  $22^\circ$ . Find its height.